You are an expert in statistical methodology and mathematical reasoning. Your task is to produce a full formal proof for a problem in an ongoing research project.

Problem: Prove that for a copula C on $[0,1]^2$ and parameters $\gamma > 0$, $\delta \in (0,2)$, define

$$\kappa_L(\delta, \gamma; C) = \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta}, \gamma^{-1} u^{2-\delta}\big)}{\log u},$$

whenever the limit exists. The claim to be proved is that under suitable regularity conditions, $\kappa_L(\delta, \gamma; C)$ does not depend on the choice of scaling parameter γ for all $\delta \in (0,2)$. In addition, verify that the proposed copula C_{PM} described in Context also satisfies this claim.

Context: We propose a new class of copulas that can account for angular full-range tail dependence, capturing not only asymmetry but also non-exchangeability. Our proposed copula, denoted by C_{PM}, is defined as following. Consider a random pair:

$$X_1 = \frac{R}{R^*} \times \frac{H_1}{H_1^*}, \qquad X_2 = \frac{R}{R^*} \times \frac{H_2}{H_2^*}, \tag{1}$$

where Pa denotes Pareto distribution, $R \sim \mathrm{Pa}(\beta)$, $R^* \sim \mathrm{Pa}(\alpha)$, $H_1 \sim \mathrm{Pa}(b_1)$, $H_1^* \sim \mathrm{Pa}(a_1)$, $H_2 \sim \mathrm{Pa}(b_2)$, $H_2^* \sim \mathrm{Pa}(a_2)$, with all shape parameters α, β, a_1, a_2, b_1, $b_2 > 0$. Moreover, these factor RVs R, R^*, H_1, H_2, H_1^*, H_2^*, are mutually independent. The copula underlying the random pair (X_1, X_2), which we term the Pareto-mixture copula, is denoted by C_{PM} and can be computed via

$$C_{\mathrm{PM}}(u_1, u_2) = F_{X_1,X_2}\big(F_{X_1}^{-1}(u_1), F_{X_2}^{-1}(u_2)\big), \qquad u_1, u_2 \in [0,1], \tag{2}$$

where F^{-1} denote the inversion of a CDF F. To theoretically analyze this new model class, we first need a principled measure of tail behavior that can vary with the direction (angle) of approach to the tail. That is the motivation of how we define $\kappa_L(\delta, \gamma; C)$. This quantity is a generalized lower-tail dependence coefficient (Anthony W Ledford and Jonathan A Tawn. Statistics for near independence in multivariate extreme values. Biometrika, 83(1):169–187, 1996.). Our intuition is that, when the limit exists, the resulting angular tail behavior should primarily be governed by the direction parameter δ, and should not depend on the scaling parameter γ. We therefore seek (i) regularity conditions under which such scaling invariance holds and (ii) a rigorous proof under those conditions.

Given the problem and context above, give a full formal proof for the problem. State any assumptions that are needed. Justify each step and give the complete proof in LaTeX.